

Transistor Switching Network Synthesis

Modeling, State Representation, and Generation Process

A formal introduction to switching networks,
Boolean function representation, and the sequential
generation MDP with SOP-based state initialization.

TransNetGame Project

Contents

1	Switching Networks	2
1.1	Basic Definitions	2
1.2	Correctness and Safety	2
1.3	Monotonicity — A Key Structural Property	2
2	SOP Representation and the Series-Parallel Network	3
2.1	Sum-of-Products (SOP)	3
2.2	Series Chain per Minterm	3
2.3	Parallel Combination: the SOP Series-Parallel Network	3
2.4	Illustrated Example: XOR ($N = 2$)	3
3	Sequential Generation as a Markov Decision Process	4
3.1	Two-Component State	4
3.2	The Compensate Network: Always the Full SOP	4
3.3	Action Space	5
3.4	Transition	5
3.5	Termination	6

1. Switching Networks

1.1 Basic Definitions

We work with N Boolean input variables $\mathbf{x} = (x_1, \dots, x_N)$. An input pattern $\pi \in \{0, 1\}^N$ assigns a concrete value to each variable. The **literal alphabet** is

$$\mathcal{L} = \{x_1, \neg x_1, x_2, \neg x_2, \dots, x_N, \neg x_N\}, \quad |\mathcal{L}| = 2N.$$

Literal x_j evaluates to π_j under π ; literal $\neg x_j$ evaluates to $1 - \pi_j$.

Definition 1.1 (Switching Network). A **switching network** over variables $\mathbf{x}$ is an undirected multigraph

$$G = (V, E), \quad V \supseteq \{\mathbf{S}, \mathbf{T}\}, \quad E \subseteq \binom{V}{2} \times \mathcal{L},$$

where $\mathbf{S}$ (source) and $\mathbf{T}$ (drain/sink) are two distinguished terminal nodes. Every edge $(u, v, \ell) \in E$ represents a **transistor** with control literal $\ell \in \mathcal{L}$.

Definition 1.2 (Conduction). Under input pattern π , transistor (u, v, ℓ) **conducts** if and only if $\ell(\pi) = 1$. Network G **conducts** $\mathbf{S} \rightarrow \mathbf{T}$ under π if and only if there exists a path from $\mathbf{S}$ to $\mathbf{T}$ in the subgraph of G restricted to conducting transistors.

1.2 Correctness and Safety

Given a Boolean function $f : \{0, 1\}^N \rightarrow \{0, 1\}$, partition all 2^N input patterns into:

$$P^+ = \{\pi : f(\pi) = 1\} \quad (\text{on-patterns}), \quad P^- = \{\pi : f(\pi) = 0\} \quad (\text{off-patterns}).$$

Definition 1.3 (Correct Implementation). Network G **correctly implements** f if and only if:

$$\begin{aligned} \forall \pi \in P^+ : \quad & \mathbf{S} \rightarrow \mathbf{T} \text{ conducts in } G \text{ under } \pi, & (\text{Correctness}) \\ \forall \pi \in P^- : \quad & \mathbf{S} \not\rightarrow \mathbf{T} \text{ in } G \text{ under } \pi. & (\text{Safety}) \end{aligned}$$

The synthesis goal is to find a correct implementation with the minimum number of transistors:

$$\min\{|E| : G = (V, E) \text{ correctly implements } f\}.$$

1.3 Monotonicity — A Key Structural Property

Theorem 1.4 (Monotonicity of Conducting Paths). *For any network G and any transistor $e \notin E$,*

$$\text{Paths}(G \cup \{e\}) \supseteq \text{Paths}(G),$$

where $\text{Paths}(H)$ denotes the set of $(\mathbf{S}, \mathbf{T})$ conducting paths of H under all patterns.

Proof. Adding a transistor can activate new paths but cannot deactivate any existing conducting transistor. Hence every path that existed in G still exists in $G \cup \{e\}$. $\square$

Corollary 1.5 (Inheritable Safety). *If G is safe (no $\mathbf{S} \rightarrow \mathbf{T}$ path under any $\pi \in P^-$), then $G \cup \{(u, v, \ell)\}$ is safe if and only if (u, v, ℓ) does not create any $\mathbf{S} \rightarrow \mathbf{T}$ path under any $\pi \in P^-$.*

Corollary 1.5 means safety is an *incrementally checkable invariant*: we need only verify the new transistor, not the entire network.

2. SOP Representation and the Series-Parallel Network

2.1 Sum-of-Products (SOP)

Any Boolean function f can be expressed as a disjunction of **minterms**. For N variables, each on-pattern $\pi = (b_1, \dots, b_N) \in P^+$ uniquely corresponds to the product term

$$T_\pi = \ell_1^\pi \cdot \ell_2^\pi \cdots \ell_N^\pi, \quad \ell_j^\pi = \begin{cases} x_j & \text{if } b_j = 1, \\ \neg x_j & \text{if } b_j = 0. \end{cases}$$

The full SOP is $f = \bigvee_{\pi \in P^+} T_\pi$.

2.2 Series Chain per Minterm

Product term T_π maps directly to a **series chain** of N transistors. Introduce $N - 1$ fresh internal nodes $c_{\pi,1}, \dots, c_{\pi,N-1}$ and connect:

$$\mathbf{S} \xrightarrow{\ell_1^\pi} c_{\pi,1} \xrightarrow{\ell_2^\pi} c_{\pi,2} \cdots c_{\pi,N-1} \xrightarrow{\ell_N^\pi} \mathbf{T}.$$

This chain conducts under pattern π' if and only if every literal ℓ_j^π evaluates to 1 under π' , which happens precisely when $\pi' = \pi$.

2.3 Parallel Combination: the SOP Series-Parallel Network

Definition 2.1 (SOP Network). The **SOP series-parallel network** for f is the union of all minterm chains sharing terminal nodes $\mathbf{S}$ and $\mathbf{T}$:

$$G_{\text{SOP}}(f) = \left(\{\mathbf{S}, \mathbf{T}\} \cup \{c_{\pi,j} : \pi \in P^+, j = 1, \dots, N-1\}, \bigcup_{\pi \in P^+} \text{chain}(\pi) \right).$$

Theorem 2.2 (Correctness and Safety of G_{SOP}). $G_{\text{SOP}}(f)$ correctly implements f .

Proof. Correctness. For $\pi \in P^+$, every literal ℓ_j^π evaluates to 1 under π , so the chain for π is fully conducting: $\mathbf{S} \rightarrow \mathbf{T}$.

Safety. For $\pi' \in P^-$, consider any chain indexed by $\pi \in P^+$. Since $\pi' \neq \pi$, they differ in at least one bit position j^* . Then $\ell_{j^*}^{\pi'}(\pi') = 0$: the j^* -th transistor of the π -chain is off. Every chain contains at least one non-conducting transistor under π' , so no $\mathbf{S} \rightarrow \mathbf{T}$ path exists. $\square$

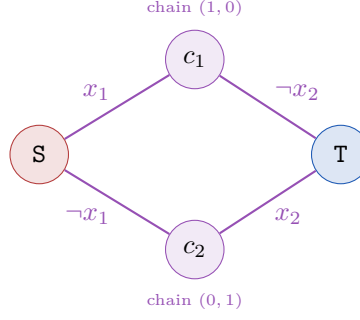
The SOP network is always *correct and safe by construction*, but is generally *not minimal*.

Quantity	Value
Transistors $ E_{\text{SOP}} $	$ P^+ \times N$
Internal nodes	$ P^+ \times (N - 1)$

2.4 Illustrated Example: XOR ($N = 2$)

$$f(x_1, x_2) = x_1 \oplus x_2, \quad P^+ = \{(1, 0), (0, 1)\}, \quad P^- = \{(0, 0), (1, 1)\}.$$

G_{SOP} consists of two series chains in parallel:



$|E_{\text{SOP}}| = 4$ transistors, $|P^+| \times N = 2 \times 2 = 4$. (Already optimal for XOR with $N = 2$.)

3. Sequential Generation as a Markov Decision Process

We model transistor network synthesis as a sequential decision process. At each step the agent places one transistor; the episode terminates when the synthesized network already covers all on-patterns.

3.1 Two-Component State

The state at every time step t consists of two disjoint subnetworks that together form the **unified graph** GC_t :

Component	Definition	Role
Synthesized network G_t	$(V_{G,t}, E_{G,t}); V_{G,0} = \{S, T\}, E_{G,0} = \emptyset$	Transistors placed by the agent so far
Compensate network C_t	$G_{\text{SOP}}(f)$; <i>fixed for all t</i>	Full SOP network for f ; encodes all on-patterns
Unified graph GC_t	$G_t \cup C_t = (V_{G,t} \cup V_C, E_{G,t} \cup E_C)$	Full observation available to the policy

S and T are shared between both components. Internal nodes of G_t and C_t are disjoint sets.

3.2 The Compensate Network: Always the Full SOP

At every step t , the compensate network is the *complete, fixed* SOP series-parallel network for the entire Boolean function f :

Definition 3.1 (Compensate Network). The compensate network is the full SOP series-parallel network for f and is *fixed throughout the entire episode*:

$$C_t = G_{\text{SOP}}(P^+) = G_{\text{SOP}}(f) \quad \text{for all } t \geq 0.$$

It contains one series chain per on-pattern in P^+ , and all chains share the terminals S and T.

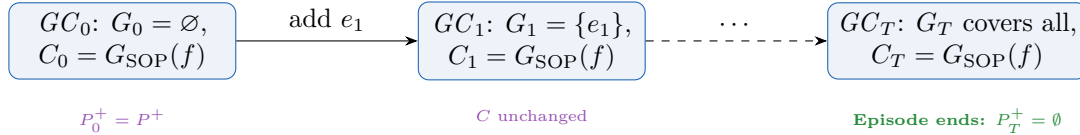
Why always the full SOP?

The compensate network is a *fixed reference* that encodes the complete specification of the Boolean function throughout the episode.

- **Information completeness:** C_t always encodes *every* on-pattern as an explicit series chain; the policy always has full knowledge of the function to be synthesized.

- **Structural stability:** The same series-parallel topology is present at every step, giving the policy a consistent graph structure regardless of how many transistors have been placed.
- **No Boolean minimization:** Building $G_{\text{SOP}}(f)$ requires only enumerating P^+ ; no external logic solver is needed.

The compensate network $C_t = G_{\text{SOP}}(f)$ remains unchanged from step 0 to termination. Coverage progress is tracked separately via $P_t^+ \subseteq P^+$ (on-patterns not yet routed in G_t). The episode ends when $P_t^+ = \emptyset$, i.e. G_t alone covers all on-patterns.



3.3 Action Space

Each action is a triple

$$a_t = (\text{src}, \text{drn}, \ell) \in V_{G,t} \times (V_{G,t} \cup \{n_{\text{new}}\}) \times \mathcal{L},$$

where n_{new} denotes a fresh internal node to be instantiated. The action is subject to two structural constraints enforced before any literal is selected:

Canonical ordering (symmetry breaking). Transistors are undirected, so (u, v, ℓ) and (v, u, ℓ) are identical. To eliminate duplicates, the drain must satisfy:

$$\text{valid_drn}(\text{src}) = \{v \in V_{G,t} \cup \{n_{\text{new}}\} : v = \mathbf{T} \text{ or } \text{idx}(v) > \text{idx}(\text{src})\}.$$

$\mathbf{T}$ is always a valid drain regardless of its index.

Safety mask (BFS cutoff). Once (src, drn) is fixed, any literal whose addition would create a $\mathbf{S} \rightarrow \mathbf{T}$ path under any $\pi \in P^-$ is forbidden:

$$\mathcal{F}(\text{src}, \text{drn}) = \{\ell \in \mathcal{L} : \exists \pi \in P^-, \mathbf{S} \rightarrow \mathbf{T} \text{ in } (V_{G,t}, E_{G,t} \cup \{(\text{src}, \text{drn}, \ell)\}) \text{ under } \pi\}.$$

Only literals in $\mathcal{L} \setminus \mathcal{F}(\text{src}, \text{drn})$ are allowed.

3.4 Transition

Given action $a_t = (\text{src}, \text{drn}, \ell)$:

1. **Update synthesized network.** If $\text{drn} = n_{\text{new}}$, create a new node n_k and set $\text{drn} \leftarrow n_k$:

$$E_{G,t+1} = E_{G,t} \cup \{(\text{src}, \text{drn}, \ell)\}, \quad V_{G,t+1} = V_{G,t} \cup \{\text{drn}\}.$$

2. **Coverage check on G_{t+1} only.** Find newly covered on-patterns:

$$\Delta_t = \{\pi \in P_t^+ : \mathbf{S} \rightarrow \mathbf{T} \text{ in } G_{t+1} \text{ under } \pi\}.$$

3. **Update coverage tracking.** Record newly covered on-patterns:

$$P_{t+1}^+ = P_t^+ \setminus \Delta_t.$$

The compensate network is *not* modified: $C_{t+1} = G_{\text{SOP}}(f)$.

4. **New unified state.** $GC_{t+1} = G_{t+1} \cup C_{t+1}$.

Implementation note. The BFS safety check (step 2 of the action selection) must use $E_{G,t}$ only — not the combined $E_{G,t} \cup E_{C,t}$. Compensate transistors represent coverage obligations, not physical switches in the circuit. The implementation must therefore maintain $E_{G,t}$ and $E_{C,t}$ as separate edge sets, even though GC_t presents them as a single unified graph to the policy.

3.5 Termination

The episode terminates when $P_t^+ = \emptyset$, i.e. when G_t routes $\mathbf{S} \rightarrow \mathbf{T}$ under every $\pi \in P^+$. The compensate network $C_t = G_{\text{SOP}}(f)$ is present but no longer contributes new obligations. At that point:

- **Correctness:** G_T routes $\mathbf{S} \rightarrow \mathbf{T}$ under every $\pi \in P^+$ (since $P_T^+ = \emptyset$ means all patterns were covered during the episode).
- **Safety:** G_T never routes $\mathbf{S} \rightarrow \mathbf{T}$ under any $\pi \in P^-$ (guaranteed by the BFS mask at every step, by induction from Corollary 1.5).

No explicit “stop” action is required.